ChatGPT

Query-relative structure, phenomenology, and structural realism: foundations for a non-equilibrium statistical framework

Overview and conceptual map

A "query-relative" statistical framework treats **observable regularities** as indexed by a family of **queries** (measurement/observation protocols), and treats any global "law" (e.g., a path-space probability measure) as a **representation** that is admissible only if it coherently realizes those query-indexed regularities. This stance is naturally **pre-dynamical**: it asks whether multi-resolution summaries are mutually extendable before asking whether they arise from a Markov kernel, semigroup, generator, etc. (Mathematically, this parallels the move from *assuming* a process to *testing* for projective/extendable consistency.) [1]

This architecture is well-suited to act as a bridge among four large literatures:

- **Phenomenology**: emphasizes that experience is perspectival yet structured (a "horizon" with stable invariants rather than atomistic sensations). [2]
- **Structural realism**: argues that science's durable commitments concern *structure/relations* more than intrinsic "natures" of unobservable entities. [3]
- **Information geometry**: characterizes what is invariant under "changes of perspective" formalized as sufficient statistics / Markov morphisms; uniquely singled out structures (e.g., Fisher metric) are invariants of such transformations. [4]
- **Non-equilibrium statistical mechanics**: is already, in many modern formulations, a theory of **path-space structure**, **coarse-graining**, and **divergences** (e.g., relative entropy) quantifying irreversibility and information flow. [5]

A key payoff is philosophical: a query-first completion notion can avoid both (i) naïve "raw data realism" and (ii) strong metaphysical realism about intrinsic dynamical structure, by making **global dynamics an admissible representation** constrained by cross-query compatibility, rather than a primitive. [6]

Phenomenology: horizon, embodiment, and invariance in experience

Maurice Merleau-Ponty [7] 's mature phenomenology is widely read as rejecting two extremes: (a) atomistic empiricism (experience as a mosaic of "sensations") and (b) pure intellectualism (structure imposed by detached reason). The core claim is that perception is **primarily embodied** and already comes with a **field-structure**—a pre-reflective organization of figure/ground, affordances, and temporal flow—rather than assembling "data points" into a world. The standard scholarly entry-point is the overview of his method and its development in his main works. [8]

Phenomenologists often use “horizon” language to express that what is given is always given **against** a background of implicit possibilities, dependencies, and orientations. The point for a query-relative framework is not to import phenomenology as metaphysics, but to notice a structural analogy: **each query** (each way of “taking up” the phenomenon) yields a partial profile, while coherence across queries is what stabilizes a higher-order object (“the same thing” across perspectives). This is precisely the role played by extension/completion conditions in a query-first probability framework. [9]

A particularly relevant bridge-text for your project is Philipp Berghofer [10] ’s work on scientific perspectivism in the phenomenological tradition. It explicitly frames Merleau-Ponty as occupying a position “between” standard realism and anti-realism, and argues that Merleau-Ponty’s “partial realism” can be understood as prioritizing **structural relations** (mathematical forms describing subject–object relations) over intrinsic observer-independent objects. This is a direct conceptual match to a “horizon of query-regularities” constraining admissible global representations. [11]

Within contemporary embodied and enactive traditions, the link between phenomenology and scientific formalization is often pursued via sensorimotor coupling and “bringing forth” domains of significance. The Embodied Mind [12] is a canonical reference, positioning embodied cognition as a route for connecting phenomenological description to scientific modeling (without reducing one to the other). [13]

Structural realism: epistemic vs ontic variants and the role of observation

Structural realism is frequently presented as a selective realist response to the tension between (i) the “no miracles” motivation for realism and (ii) the pessimistic meta-induction from theory change. The contemporary locus classicus for this reintroduction is John Worrall [14] ’s 1989 “best of both worlds” proposal, arguing that what gets preserved across major theory change is often mathematical/relational structure rather than theoretical ontology (e.g., Fresnel → Maxwell in optics). [15]

The Stanford Encyclopedia overview (authored by James Ladyman [16]) distinguishes **epistemic structural realism (ESR)** from **ontic structural realism (OSR)** and tracks how structural realism has been motivated by both historical cases and considerations from modern physics. It also notes key ancestral influences (commonly including Henri Poincaré [17]) and clarifies the role of relational structure in OSR-style views. [18]

A standard analytic touchstone on the ESR/OSR distinction is Ladyman’s 1998 paper “What is structural realism?”, which frames structural realism as (i) avoiding overcommitment to “furniture” while (ii) retaining the explanatory force of realism by treating structure as world-connecting. [19]

More radical OSR programs are prominently developed in Every Thing Must Go: Metaphysics Naturalized [20] (Don Ross [21] coauthor), which argues for a metaphysics disciplined by contemporary science and defends OSR as a candidate for that metaphysics. [22]

A book-length OSR defense and development is The Structure of the World: Metaphysics and Representation [23] by Steven French [24] , which explicitly places structure (often symmetries and laws) at the center of ontology and scientific representation. [25]

Why structural realism is especially compatible with a query-first boundary framework: structural realism already says that what is robustly knowable is **relational structure**, often characterized by what survives transformation/translation between theoretical perspectives. A query-first completion notion makes this precise in an operational-probabilistic way: the "structure" is what is **stable and coherent across a family of admissible queries**, while any deeper "interior ontology" is underdetermined by (and therefore not compelled by) the observable structure.

Query-relative compatibility: probability-theoretic and sheaf-theoretic formalisms

Two existing mathematical traditions provide highly relevant templates for "completion from perspectives" without assuming a pre-given whole.

Projective consistency and Kolmogorov-style existence

The classical existence theorem for processes starts from families of finite-dimensional distributions satisfying consistency conditions and concludes that there exists a measure on an infinite product space realizing them. In its basic form, "if a family of measures fits together consistently, it defines a measure on the infinite product." [26]

This is structurally aligned with your boundary completion stance:

- **Queries** ↔ selecting finite coordinate sets / finite-resolution observables
- **Observed regularities** ↔ finite-dimensional marginals
- **Completion** ↔ existence of a global path-space measure (when it exists)

Crucially, the theorem is not metaphysical: it does not assert that the global measure is "the reality"; it asserts that (under conditions) a coherent global representation exists. The global object is a **witness of coherence**. [26]

Contextuality and "global section" obstructions (sheaf-theoretic viewpoint)

A different but strikingly parallel template comes from the sheaf-theoretic analysis of nonlocality/contextuality. Here, one begins with a family of empirical probability assignments on overlapping measurement contexts and asks whether there exists a **single global assignment** consistent with all overlaps. Samson Abramsky [27] and Adam Brandenburger [28] show that contextuality corresponds to obstructions to the existence of global sections in an associated sheaf/presheaf structure. [29]

This provides language for your phenomenon of "completion failure":

- A boundary horizon specified by query-indexed regularities is like a presheaf of local distributions.
- An exact completion is like a global section.
- Failure of completion is not merely "lack of data"; it can be a **structural obstruction**—an incompatibility among perspectives. [29]

Even if your target domain is classical statistical mechanics rather than quantum contextuality, the mathematics of "patching local marginals into a global object" is shared, and the sheaf-theoretic language is valuable because it treats incompatibility as an intrinsic, diagnosable property rather than an epistemic defect.

Information geometry: invariance under sufficiency as commensurability of perspectives

A query-first framework needs a way to compare regularities across resolutions and protocols. Information geometry offers a principled invariance story: treat transformations between observed statistical models as "changes of perspective" and ask what geometric structures are invariant.

A central modern formulation is that the Fisher metric (and related Amari–Chentsov tensor structures) can be characterized by invariance under sufficient statistics / Markov morphisms. In a rigorous generalization of Čencov's classical result, Ay and collaborators show that the Fisher metric and Amari–Chentsov tensor are (under suitable conditions) uniquely characterized—up to scale—by invariance under sufficient statistics, and they analyze Markov morphisms between statistical models in these terms. [30]

This is directly relevant to your question "Can divergences and information geometry be interpreted as commensurability measures between perspectives?":

- A **query** induces a pushforward of a more detailed regularity to a coarser outcome space (statistic / Markov map).
- **Sufficiency** is the condition that this mapping loses no relevant information for a task/model class.
- An "information-geometric invariant" is then a structure that does not depend on the particular representative of the perspective, only on what is preserved across sufficient transformations. [31]

In this light, "information" in a query-first framework is naturally understood less as a substance ("bits in the world") and more as **invariance-class structure**: what remains stable under admissible transformations of viewpoint/protocol. This resonates strongly with structural realism's focus on preserved structure and with phenomenology's emphasis on invariants across profiles of appearance—without reducing either to the other.

Non-equilibrium statistical mechanics: path measures, divergences, coarse-graining

Modern non-equilibrium theory is, in many places, explicitly a theory of **path-space regularities** and their transformations under observation and coarse-graining.

Trajectory-level structure and fluctuation relations

Udo Seifert [32]'s influential review frames stochastic thermodynamics as assigning thermodynamic quantities (work, heat, entropy production) at the level of individual trajectories and deriving integral/ detailed fluctuation theorems that constrain trajectory-dependent observables under non-equilibrium driving and steady states. [33]

Canonical results include:

- Christopher Jarzynski [34] 's equality relating equilibrium free energy differences to an exponential average of non-equilibrium work over trajectories. [35]
- Gavin E. Crooks [36] ' fluctuation theorem relating forward and reverse work distributions (and providing a route to Jarzynski-type relations). [37]

These results are important for your program not just as "physics facts," but because they already encode the idea that observable constraints on trajectories can be formulated in terms of **probability measures on path space** and their relations under reversal and coarse-graining.

Entropy production as distinguishability of forward vs reverse

A particularly direct bridge from non-equilibrium physics to your divergence-based completion viewpoint is the line of work linking entropy production to relative entropy (KL divergence) between forward and time-reversed descriptions.

- Pierre Gaspard [38] develops a path-entropy approach in which entropy production in Markovian random processes is related to differences between forward and time-reversed dynamical entropies, and explicitly ties these differences to relative entropy rates. [39]
- Juan M. R. Parrondo [40] , Christian van den Broeck [41] , and Ryoichi Kawai [42] present an explicit relationship between entropy production and distinguishability from the time-reversed process, quantified via relative entropy. [43]

This is almost tailor-made for a query-first framework: divergences are not just "fit measures"; they become physical measures of irreversibility/time-asymmetry under a chosen observational regime.

Large deviations as the statistical mechanics of path-space

Hugo Touchette [44] 's review positions large deviation theory as providing the mathematical language underlying many variational principles and fluctuation structures in statistical mechanics, including nonequilibrium settings and Markov processes. [45]

For a query-first non-equilibrium framework, large deviations contribute two foundational ideas:

1. **Multi-scale closure as a variational principle:** the "best" effective description at a coarser level often arises from a constrained minimization problem (frequently involving relative entropy or rate functions). [46]
2. **Empirical regularities as exponential-scale structure:** what matters is not only whether a completion exists, but the rate at which incompatibilities manifest or vanish with sample size/time (a natural role for your approximate completion tolerances). [47]

A complementary modern synthesis of entropy production across classical and quantum settings is given in the review by G. T. Landi [48] and collaborators, emphasizing entropy production as a unifying quantity in finite-time thermodynamics. [49]

Neutrality, underdetermination, and admissible interior realizations

A central requirement you stated is that the framework should leave "enough space" for different philosophical readings—empiricist, rationalist, phenomenological, realist, instrumentalist—without baking in a dogma. The literature above suggests a precise way to do this:

- Treat the query-indexed regularities as **primitive constraints** (phenomenology-friendly: the "field" of appearance is structured; structural-realist-friendly: structure is what is preserved). [6]
- Treat global dynamics (path measures, Markov semigroups, generators) as **representational posits** whose admissibility is conditional on cross-query consistency (probability-theoretic completion) and additional regularity (e.g., semigroup properties, continuity). [50]

Two points from the philosophical literature reinforce the neutrality you want:

1. **Merleau-Ponty as neither instrumentalist nor constructivist:** Berghofer explicitly notes that Merleau-Ponty's "partial realism" is intended to be neither traditional anti-realism nor a denial that science can tell us about reality, while still rejecting observer-independent intrinsic objects as the fundamental ontological units. This supports your desire to avoid collapsing into either British-style "raw givens" empiricism or a pre-given rationalist whole. [51]

2. **Structural realism's built-in underdetermination:** structural realism is often motivated precisely because it can respect theory change and underdetermination: multiple incompatible ontologies may realize the same preserved structure. Your query-first completion notion formalizes that underdetermination as "multiple admissible completions/realizations of the same boundary horizon." [52]

A physics-side analogue of this perspectival/relational stance appears in Carlo Rovelli [53]'s relational quantum mechanics, which argues that observer-independent "states" are not the right primitive and that physical description is relational (information between systems) rather than absolute. Whether or not one accepts RQM, it illustrates how a perspectival stance can remain realist-leaning while rethinking what "objective structure" is. [54]

Putting this together: the most defensible "neutral" claim your framework can make—consistent with both the structural realist and phenomenological literatures—is:

> The observable horizon (query-indexed regularities) constrains the class of admissible global representations; it need not uniquely determine them, and it need not presuppose a pre-given intrinsic dynamical law.

This is simultaneously: - structurally realist (structure is what survives across perspectives), [55]
- phenomenology-compatible (the "whole" is stabilized through coherent profiles, not reducible to any one profile), [56]
- and non-equilibrium-native (path-space divergences, coarse-graining, and fluctuation structures already operationalize these ideas). [57]

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